

Found 20947 images belonging to 10 classes.

Found 5232 images belonging to 10 classes.

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_5 (Conv2D)	(None, 55, 55, 96)	34,944
batch_normalization_3 (BatchNormalization)	(None, 55, 55, 96)	384
max_pooling2d_3 (MaxPooling2D)	(None, 27, 27, 96)	0
conv2d_6 (Conv2D)	(None, 27, 27, 256)	614,656
batch_normalization_4 (BatchNormalization)	(None, 27, 27, 256)	1,024
max_pooling2d_4 (MaxPooling2D)	(None, 13, 13, 256)	0
conv2d_7 (Conv2D)	(None, 13, 13, 384)	885,120
conv2d_8 (Conv2D)	(None, 13, 13, 384)	1,327,488
conv2d_9 (Conv2D)	(None, 13, 13, 256)	884,992
batch_normalization_5 (BatchNormalization)	(None, 13, 13, 256)	1,024
max_pooling2d_5 (MaxPooling2D)	(None, 6, 6, 256)	0
flatten_1 (Flatten)	(None, 9216)	0

dense_3 (Dense)	(None, 4096)	37,752,832
dropout_2 (Dropout)	(None, 4096)	0
dense_4 (Dense)	(None, 4096)	16,781,312
dropout_3 (Dropout)	(None, 4096)	0
dense_5 (Dense)	(None, 10)	40,970

Total params: 58,324,746 (222.49 MB)

Trainable params: 58,323,530 (222.49 MB)

Non-trainable params: 1,216 (4.75 KB)

None

Epoch 1/2

655/655 ————— **1771s** 3s/step - accuracy: 0.2951 - loss: 2.7817 - val_accuracy: 0.3979 - val_loss: 1.7693

Epoch 2/2

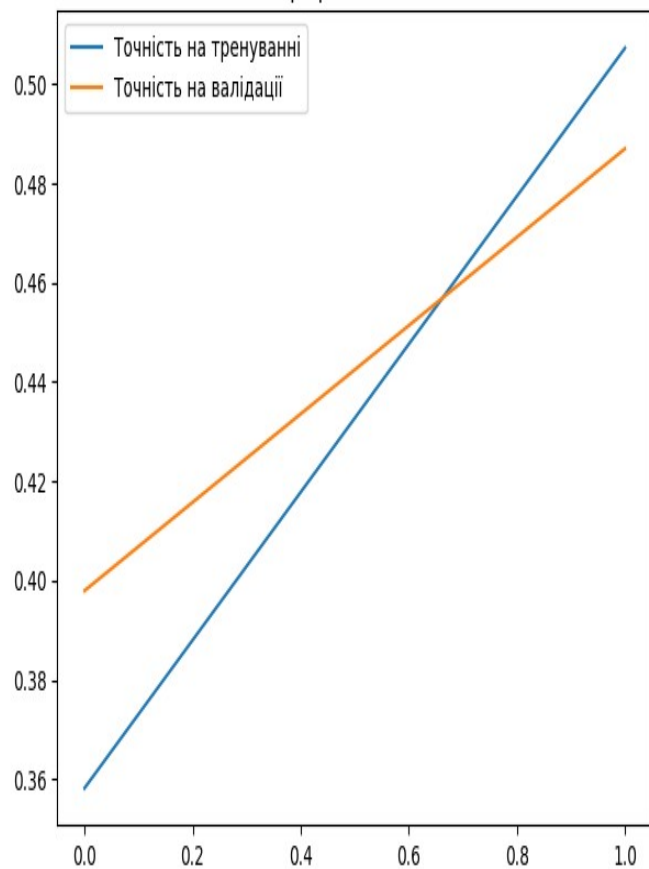
655/655 ————— **1754s** 3s/step - accuracy: 0.4957 - loss: 1.5061 - val_accuracy: 0.4870 - val_loss: 1.5739

164/164 ————— **92s** 562ms/step - accuracy: 0.4872 - loss: 1.5861

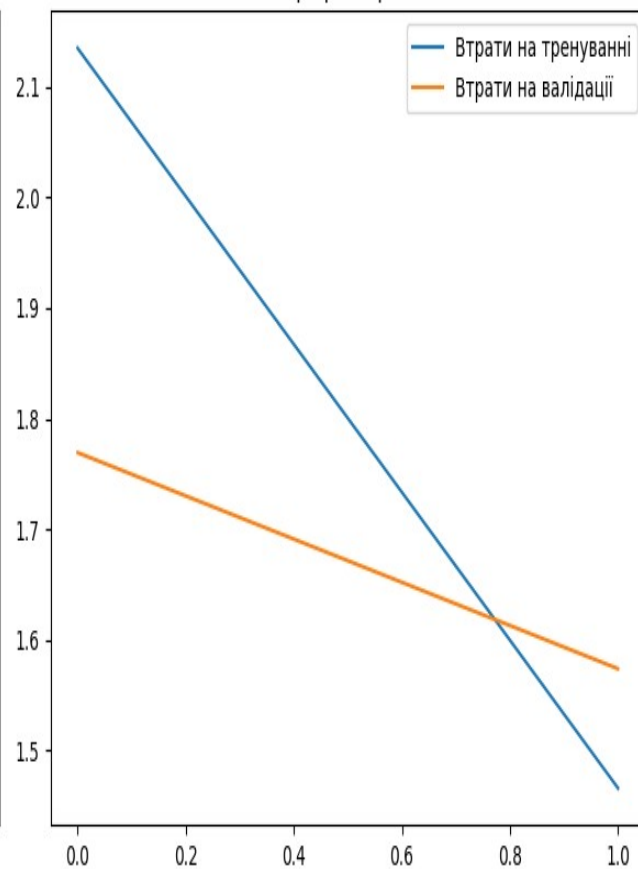
Точність на валідації: 0.4870

Втрати (Loss): 1.5739

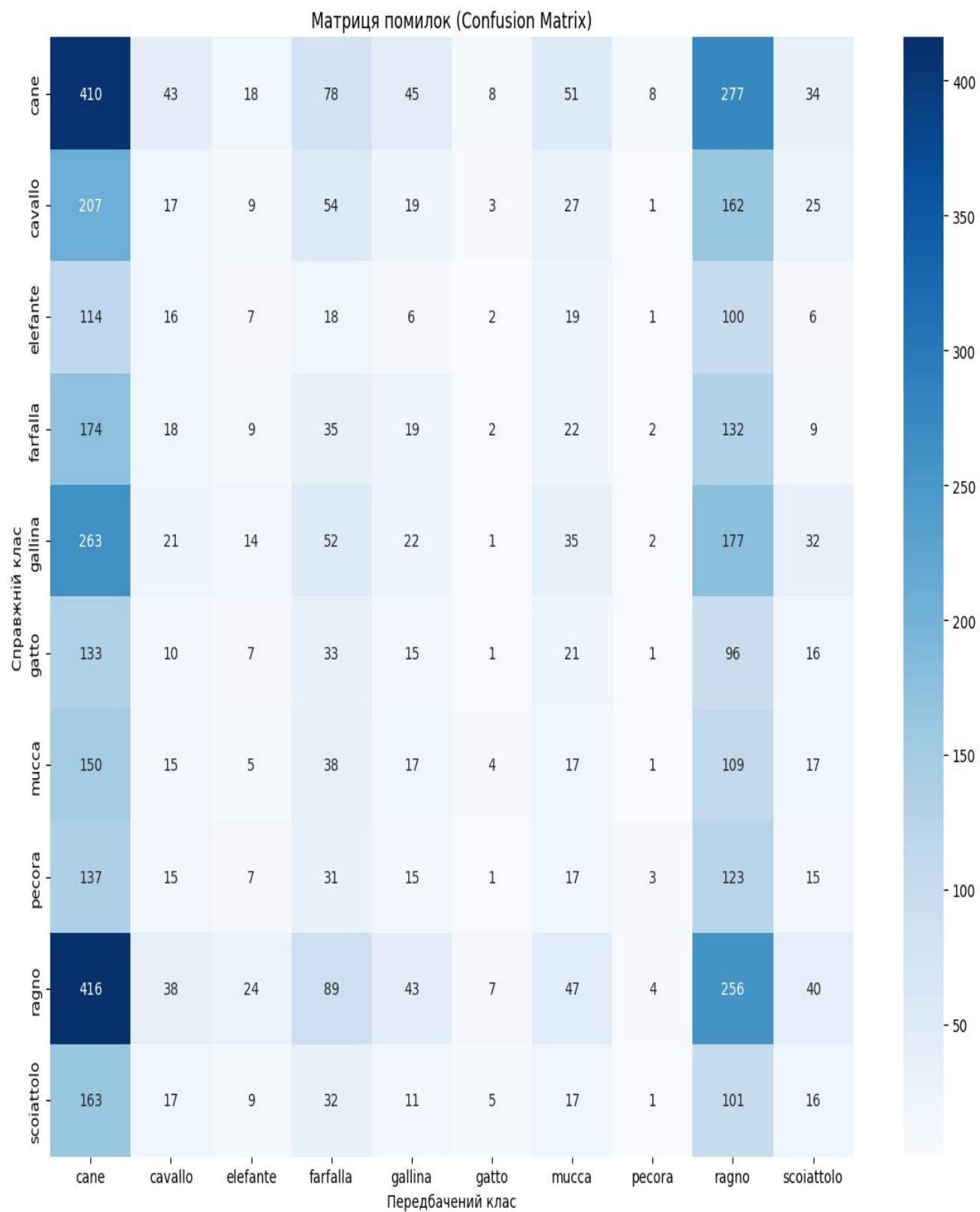
Графік точності



Графік втрат



164/164 ————— 92s 562ms/step



Accuracy : 0.1498

Precision : 0.1164

Recall : 0.1498

F-Score : 0.1146

Детальний звіт по класах:

	precision	recall	f1-score	support
cane	0.19	0.42	0.26	972
cavallo	0.08	0.03	0.05	524
elefante	0.06	0.02	0.04	289
farfalla	0.08	0.08	0.08	422
gallina	0.10	0.04	0.05	619
gatto	0.03	0.00	0.01	333
mucca	0.06	0.05	0.05	373
pecora	0.12	0.01	0.02	364
ragno	0.17	0.27	0.21	964
scoiattolo	0.08	0.04	0.05	372
accuracy		0.15		5232
macro avg	0.10	0.10	0.08	5232
weighted avg	0.12	0.15	0.11	5232

1/1  **0s** 213ms/step

[[0.05191917 0.38518173 0.10838426 0.00263582 0.0021454 0.00411802
0.4172742 0.01877295 0.00375233 0.00581611]]

Очікуваний: cavallo
Передбачений: тусса



4/4 ————— 2s 555ms/step

4/4 ————— 2s 548ms/step

Результати збережені у classification_results.csv

Шлях до

файлу Справжній клас Розпізнаний клас

		./raw-img/scoiattolo/OIP-
ADOooAvJ5Vag13ol8mDKVAHaE8.jpeg	scoiattolo	cane
		./raw-img/ragno/OIP-
GjPUDWgG3adSbLkRZvBtOwEsDh.jpeg	ragno	ragno
		./raw-img/farfalla/OIP-77GdvQVid2bO2gdlvCrx1QHaE5.jpeg
farfalla	cane	
		./raw-img/farfalla/OIP-EkFr4t-wDeNf54WHB6vDGQHaFj.jpeg
farfalla	farfalla	
		./raw-img/ragno/ea34b3072ff0073ed1584d05fb1d4e9fe777ead218ac104497f5c97ca5ecb5b1_640.jpg
ragno	ragno	
		./raw-img/pecora/eb36b60d29f4043ed1584d05fb1d4e9fe777ead218ac104497f5c978a6eab2b0_640.jpg
pecora	cane	
		./raw-img/cane/OIP-
3K1Y070zliiWG7NWZMpwPQHaE6.jpeg	cane	cane

./raw-img/ragno/OIP-

02MVkVV8gZF8y7G1plxT3AHaLh.jpeg ragno ragno

./raw-img/pecora/eb33b9092cf4013ed1584d05fb1d4e9fe777ead218ac104497f5c978a6ebb3bf_640.jpg

pecora cane

./raw-img/gallina/OIP-gyGDhH0JkukA-

kEjWT1osQHaHv.jpeg gallina cane

=== Оригінальна AlexNet ===

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_5 (Conv2D)	(None, 55, 55, 96)	34,944
batch_normalization_3 (BatchNormalization)	(None, 55, 55, 96)	384
max_pooling2d_3 (MaxPooling2D)	(None, 27, 27, 96)	0
conv2d_6 (Conv2D)	(None, 27, 27, 256)	614,656
batch_normalization_4 (BatchNormalization)	(None, 27, 27, 256)	1,024
max_pooling2d_4 (MaxPooling2D)	(None, 13, 13, 256)	0
conv2d_7 (Conv2D)	(None, 13, 13, 384)	885,120
conv2d_8 (Conv2D)	(None, 13, 13, 384)	1,327,488
conv2d_9 (Conv2D)	(None, 13, 13, 256)	884,992
batch_normalization_5 (BatchNormalization)	(None, 13, 13, 256)	1,024

max_pooling2d_5 (MaxPooling2D)	(None, 6, 6, 256)	0
flatten_1 (Flatten)	(None, 9216)	0
dense_3 (Dense)	(None, 4096)	37,752,832
dropout_2 (Dropout)	(None, 4096)	0
dense_4 (Dense)	(None, 4096)	16,781,312
dropout_3 (Dropout)	(None, 4096)	0
dense_5 (Dense)	(None, 10)	40,970

Total params: 174,971,808 (667.46 MB)

Trainable params: 58,323,530 (222.49 MB)

Non-trainable params: 1,216 (4.75 KB)

Optimizer params: 116,647,062 (444.97 MB)

=== Оптимізована AlexNet-Lite ===

Model: "AlexNet_Lite"

Layer (type)	Output Shape	Param #
conv2d_10 (Conv2D)	(None, 55, 55, 64)	23,296
batch_normalization_6 (BatchNormalization)	(None, 55, 55, 64)	256
max_pooling2d_6 (MaxPooling2D)	(None, 27, 27, 64)	0
conv2d_11 (Conv2D)	(None, 27, 27, 128)	204,928

batch_normalization_7 (BatchNormalization)	(None, 27, 27, 128)	512
max_pooling2d_7 (MaxPooling2D)	(None, 13, 13, 128)	0
conv2d_12 (Conv2D)	(None, 13, 13, 192)	221,376
conv2d_13 (Conv2D)	(None, 13, 13, 192)	331,968
conv2d_14 (Conv2D)	(None, 13, 13, 128)	221,312
batch_normalization_8 (BatchNormalization)	(None, 13, 13, 128)	512
max_pooling2d_8 (MaxPooling2D)	(None, 6, 6, 128)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 128)	0
dense_6 (Dense)	(None, 512)	66,048
dropout_4 (Dropout)	(None, 512)	0
dense_7 (Dense)	(None, 10)	5,130

Total params: 1,075,338 (4.10 MB)

Trainable params: 1,074,698 (4.10 MB)

Non-trainable params: 640 (2.50 KB)

Оригінальна AlexNet — параметрів: 58,324,746

AlexNet-Lite — параметрів: 1,075,338

Зменшення : 54.2x

Навчання AlexNet-Lite...

Epoch 1/2

655/655 — **533s** 809ms/step - accuracy: 0.3117 - loss: 2.0028 - val_accuracy: 0.3821 - val_loss: 1.7967

Epoch 2/2

655/655 — **538s** 822ms/step - accuracy: 0.5154 - loss: 1.4206 - val_accuracy: 0.2489 - val_loss: 2.6054

164/164 — **33s** 198ms/step - accuracy: 0.2435 - loss: 2.6294

AlexNet-Lite — Точність на валідації : 0.2489

AlexNet-Lite — Втрати (Loss) : 2.6054

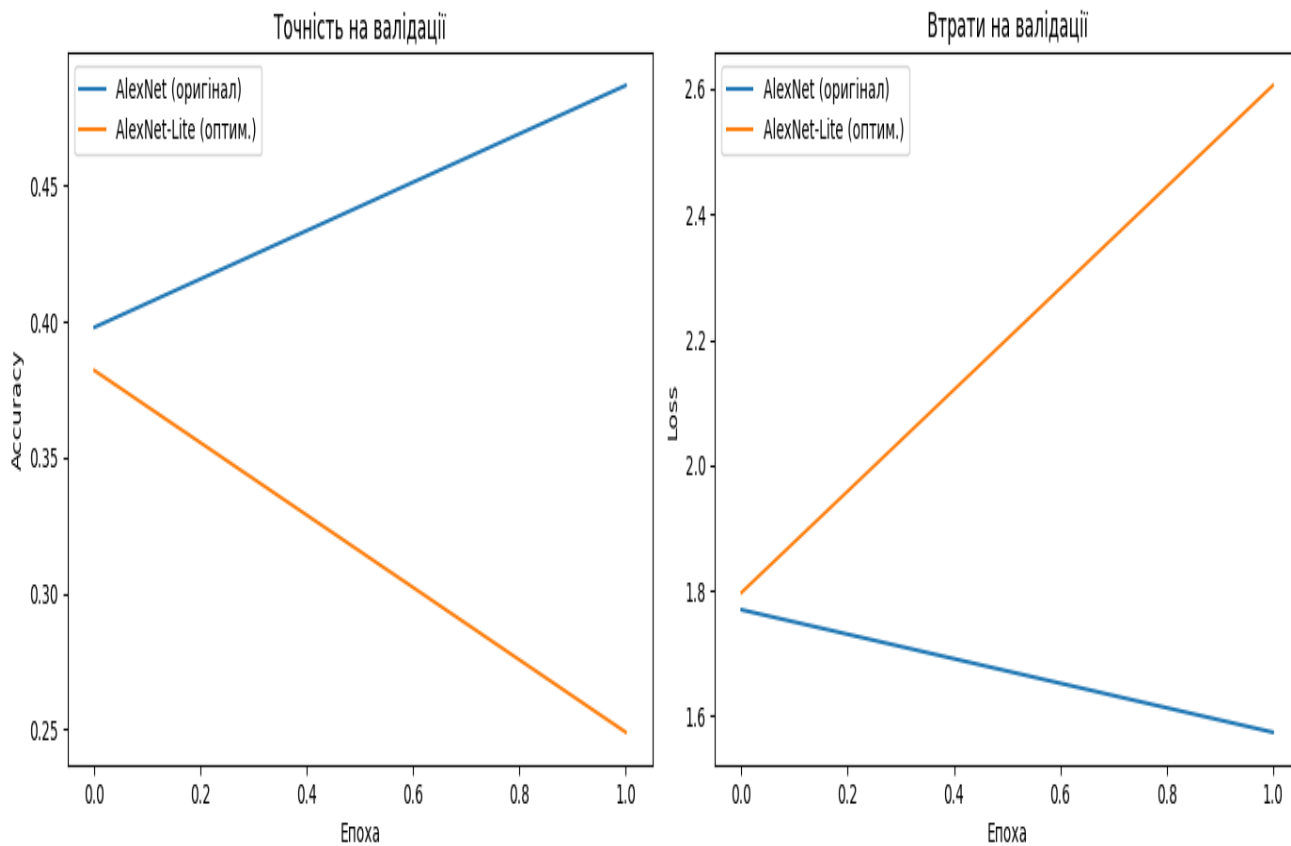
Швидкість інференсу:

Оригінальна AlexNet : 121.26 мс / зображення

AlexNet-Lite : 88.02 мс / зображення

Прискорення : 1.38x

Порівняння оригінальної AlexNet та AlexNet-Lite



=== Зведена таблиця порівняння ===

Модель Параметрів Val Accuracy Val Loss Час інференсу

AlexNet (оригінал)	58,324,746	0.4870	1.5739	121.26 мс
AlexNet-Lite (оптим.)	1,075,338	0.2489	2.6054	88.02 мс